



Perspective

Update review on SUDEP: Risk assessment, background & seizure detection devices

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ABSTRACT

This review focusses on sudden unexpected death in epilepsy patients (SUDEP) and incorporates risk stratification (through SUDEP risk factors and SUDEP risk scores), hypotheses on the mechanism of SUDEP and eligible seizure detection devices (SDDs) for further SUDEP prevention studies.

The main risk factors for SUDEP are the presence and the frequency of generalized tonic-clonic seizures (GTC). In Swedish population-based case control study, the Odds ratio of the presence of GTC in the absence of bedroom sharing is 67.

SUDEP risk scoring systems express a score that represents the cumulative presence of SUDEP risk factors, but not the exact effect of their combination. We describe 4 of the available scoring systems: SUDEP-7 inventory, SUDEP-3 inventory, SUDEP-Clinical Risk score (SUDEP-CARE score) and Kempenhaeghe SUDEP risk score. Although they all include GTC, their design is often different. Three of 4 scoring systems were validated (SUDEP-7 inventory, SUDEP-3 inventory and SUDEP-CARE score). None of the available scoring systems has been sufficiently validated for the use in a general epilepsy population.

Plausible mechanisms of SUDEP are discussed. In the MORTEMUS-study (Mortality in Epilepsy Monitoring Unit Study), SUDEP was a postictal cardiorespiratory arrest after a GTC. The parallel respiratory and cardiac dysfunction in SUDEP suggests a central dysfunction of the brainstem centers that are involved in the control of respiration and heart rhythm. In the (consequent) adenosine serotonin hypotheses SUDEP occurs when a postictal adenosine-mediated respiratory depression is not compensated by the effect of serotonin. Other (adjuvant) mechanisms and factors are discussed.

Seizure detection devices (SDDs) may help to improve nocturnal supervision. Five SDDs have been validated in phase 3 studies for the detection of TC: Seizure Link®, Epi-Care®, NightWatch, Empatica, Nelli®. They have demonstrated a sensitivity of at least 90 % combined with an acceptable false positive alarm rate. It has not yet been proven that the use will actually lead to SUDEP prevention, but clinical experience supports their effectiveness.

1. Introduction

People with epilepsy are at high risk of premature death [1]. The first studies on mortality and epilepsy (in mentally ill and retarded patients with epilepsy living in mental institutions) date from the early 20th century [2]. Epilepsy-related deaths can be classified as directly due to epilepsy (e.g., sudden unexpected death in epilepsy (SUDEP), status epilepticus or seizure-related accidents), due to acute symptomatic

seizures, indirectly due to epilepsy (e.g., aspiration pneumonia, suicide) or due to underlying neurologic disease (e.g., brain tumor, stroke) [3].

In this manuscript, we will focus on SUDEP. Since Hirsch's report of the sudden death in young epilepsy patients in 1971 [4], SUDEP was defined in 1997 as "the sudden, unexpected, witnessed or unwitnessed, non-traumatic, and non-drowning death in patients with epilepsy, with or without evidence for a seizure, with exclusion of documented status epilepticus, and when post-mortem examination does not reveal a

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structural or toxicological cause for death” [5].

Although SUDEP is a rare phenomenon, it is currently a major challenge for caregivers. Many potential risk factors for SUDEP have been investigated. The design of these studies differed substantially, and their results were sometimes contradictory. We will first review the systematic reviews and subsequently published relevant studies on SUDEP risk factors. However, for SUDEP risk stratification, these risk factors must be integrated into SUDEP risk scores based on their relevance. That is why available scores will be discussed in the second part. Next, we discuss the plausible mechanisms of SUDEP. Finally, we will study the role of SDDs in the prevention of SUDEP and we examine which seizure detection devices could be considered for clinical use and further SUDEP prevention studies.

2. Risk factors for SUDEP

2.1. Systematic reviews

Risk factors for SUDEP have been analyzed in incidence studies and in case-control studies. Controls were either living patients with epilepsy or patients with epilepsy who had died of other known causes. The relevance of each risk factor for SUDEP has been studied in systematic reviews. We searched PubMed by queries containing the terms “SUDEP” and “risk factor” up to September 17th 2023. Six hundred fifty-three papers were collected. One author (CM) screened all abstracts for systematic reviews analyzing the relevance of different risk factors for SUDEP. Six systematic reviews were included for further analysis [6–11].

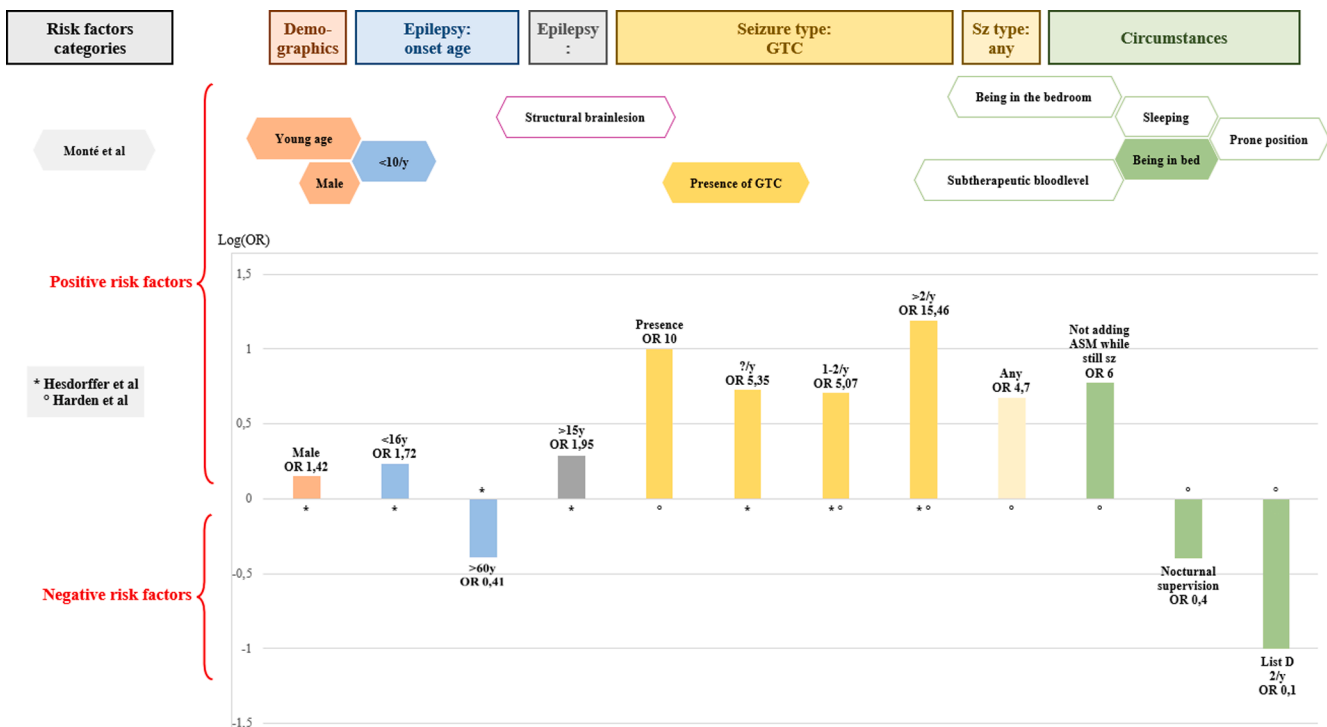
Graph 1 illustrates various important risk factors for SUDEP as stated in systematic reviews [6–11].

In a 2007 systematic review, we analyzed the relative importance of risk factors for SUDEP. For each included study, a list was used to determine which risk factors for SUDEP were investigated. Next, in

every included study we calculated a “risk factor study ratio” for every potential risk factor on our list: the square root of the ratio of the number of patients within that one study compared to the number of patients in all included studies. This risk factor study ratio was positive if the factor was considered a risk factor, negative if the factor was not considered a risk factor. It was 0 if the potential risk factor was not investigated in that particular study. The sum of all these “risk factor study ratios” was the “risk factor total studies ratio” and allowed us to rank the various risk factors for SUDEP according to their relevance. We found a number of so called “strong” risk factors for SUDEP (risk factor total studies ratio ≥ 1): young age, the presence of generalized tonic-clonic seizures (GTC), early onset of seizures, male sex and being in bed. We also found weak risk factors for SUDEP (risk factor total studies ratio ≥ 0.5 and < 1): prone position, one or more subtherapeutic blood levels of antiseizure medication, being in the bedroom, a structural brain lesion and sleeping [6].

In 2011 and 2012 the Task Force on Epidemiology of the International League Against Epilepsy (ILAE) pooled data from 4 case-control studies of risk factors for SUDEP (from the United States, Sweden, Scotland, and England) [12–15], which used living people with epilepsy as controls and used similar methodology [10,11]. The final sample consisted of 289 SUDEP cases and 958 controls (with epilepsy but without a history of heart disease) [10,11]. Sporadic GTC (1–2 per year) were associated with a more than 5-fold increase of risk (compared with not having GTC), but with frequent GTC (3 or more per year) the risk was more than 15 times higher (compared to not having GTC). The risk for SUDEP was 1.95 times higher in those patients who had had epilepsy for longer than 15 years, 1.7 times higher in those people with onset of epilepsy before the age of 16 years (vs to those with onset of epilepsy between 16 years and 60 years) and 1.4 times higher in male patients (Graph 1).

In 2017 the American Academy of Neurology (AAN) and the American Epilepsy Society (AES) pooled data [7,8] from 6 Class I and 16 Class II articles (based on AAN’s evidence base schemes to rate articles)



Graph 1. The various important risk factors for SUDEP as stated in systematic reviews: [6–11]. Onset age: <10y: younger than 10 years; <16y: younger than 16 years; >60y: older than 60 years. Epilepsy duration: >15y: more than 15 years. Seizure type: GTC: generalized tonic-clonic seizures; ?/y: unknown annual frequency; 1–2/y: 1 or 2 generalized tonic-clonic seizures a year; >2/y: at least 3 generalized tonic-clonic seizures a year; any: any seizure type. Therapy: not adding ASM while sz: Not adding antiseizure medication while still seizures. Circumstances: nocturnal supervision; List D: listening device, hexagon: Monté, *: Hesdorffer et al, °: Harden et al, OR: Odds ratio.

[10–30]. The major risk factor for SUDEP was the presence and the frequency of GTC (Graph 1). Nocturnal seizures were also considered to increase the risk in conjunction with the observation that postictal respiratory depression is a major mechanism in SUDEP. Nocturnal supervision could reduce SUDEP risk to an Odds ratio (OR) 0.4 and a remote listening device to an OR 0.1.

2.2. Studies 2017–2023

In this section, we describe relevant literature after the AAN/AES systematic review.

2.2.1. Swedish population-based case control study

In a Swedish nationwide case-control study with 255 SUDEP cases and 1148 matched controls, Sveinsson found a 67-fold increased risk of SUDEP for the combination of having at least one GTC and not sharing a bedroom with someone compared to not having GTC and sharing a bedroom (OR 67.10, 95 % CI: 29.66–151.84) [31].

In a further analysis, they concluded that frequent and nocturnal TCS carry a similar level of risk whether focal or generalized from onset. The tonic-clonic part of the seizure seems to be decisive for the fatal outcome. SUDEP risk associated with TC was the highest during the first 5 years after the epilepsy diagnosis which calls for effective TCS treatment and vigilance from the onset of diagnosis [32].

2.2.2. Incidence in children is still a matter of debate

Shortly after its publication and contrary to the AAN estimates of the 6 times lower incidence of SUDEP in children, a Swedish [31] and a Canadian study [33] reported incidence rates of definite/probable SUDEP in children being similar to rates reported in adults, whereas a later Danish study confirmed the previously described lower incidence of SUDEP in children [34]. The reason for this discrepancy is still unclear. Both Scandinavian studies [31,34] were nation-wide population-based studies and they have the same figures for adults. (Table 1) The populations were however somehow different. Keller included 17 cases of pediatric SUDEP identified through 3 sources: a national pediatrician surveillance program, child neurologist report, and screening of provincial forensic autopsies [33]. Kløvgaard included 135 cases of SUDEP in a retrospective population-based nation-wide cohort study of all Danish citizens with and without epilepsy aged 1–49 years during 2007–2009 [35].

2.2.3. Dravet syndrome

In a pooled analysis of persons with Dravet syndrome treated with fenfluramine in phase 3 clinical trials and in 2 long-term open-label observational studies, the all-cause and SUDEP mortality rates during treatment with fenfluramine was 1.7 per 1000 person-years (95 % CI: 0.4–6.7) [36]. This value was lower than the all-cause estimate of 15.8 per 1000 person-years (95 % CI: 9.9–25.4) and SUDEP estimate of 9.3 per 1000 person-years (95 % CI: 5.0–17.3) in persons with Dravet syndrome receiving standard-of-care [37]. It is unclear whether this is the result of the reduction of the seizure load or a serotonergic effect of

Table 1
SUDEP incidence in terms of age (per 1000 person-years) with 95 % confidence interval.

	AAN [7,8]	Sveinsson [31]	Keller [33]	Kløvgaard [35]
<16y	0.22	1.11	1.11	0.27
<17y	(0.16–0.31)	(0.45–2.29)	(0.63–1.79)	(0.11–0.64)
16–50y		1.13		1.21
17–49y		(0.76–1.62)		(0.96–1.51)
>50y		1.29		
		(0.88–1.82)		
Adult	1.22			
	(0.64–2.32)			

fenfluramine on the mechanism of SUDEP.

3. SUDEP risk stratification: SUDEP risk score systems

3.1. Importance

We need to move to a better prediction of SUDEP. This is relevant both for clinical purposes (e.g., to provide personalized information about the risk level and to decide whom to monitor) as well as for research (e.g., investigation of the value of preventive measures against SUDEP). An evidence based SUDEP-risk score, based on systematic reviews, should allow us to identify patients who are at higher risk for SUDEP, since it expresses the impact of the presence of one or more simultaneous risk factors for SUDEP.

3.2. Available SUDEP risk scores

We searched PubMed by queries containing the terms “SUDEP” and “score” up to September 17th 2023. Fifty-nine papers were collected. One author (CM) screened all abstracts for SUDEP risk scores, defined as a score expressing the risk for SUDEP. Ten papers were included for further analysis. We found 4 different SUDEP risk scores.

3.2.1. SUDEP-7 inventory

DeGiorgio et al. assembled the SUDEP-7 inventory (Table 2) from core risk factors from the large prospective cohort study of SUDEP (4578 patients from 3 epilepsy centers) reported by Walczak et al [12] and scored from 0 to 10 [38]. This score was initially used to analyze the correlation of heart rate variability and the risk of SUDEP [38].

3.2.2. SUDEP-3 inventory

The SUDEP-3 inventory (Table 3), was proposed to improve limitations of the SUDEP-7 inventory [39]. Tarighati Rasekhi included 28 patients who underwent video-electroencephalography (EEG) monitoring and later died of SUDEP, and 56 age- and sex-matched control patients with epilepsy. The SUDEP-7 score, its individual components, and an alternative inventory were examined. SUDEP-7 scores were significantly higher among SUDEP patients compared with controls. However, SUDEP-7 scores did not enhance the prediction of SUDEP over either GTC seizure frequency or seizure frequency alone. SUDEP-3 inventory outperformed the SUDEP-7 inventory (p = 0.01) in predicting SUDEP.

3.2.3. SUDEP-Clinical Risk score (SUDEP-CARE)

Serrand et al. conducted a case-control study including definite or probable SUDEP cases collected by the French National Sentinel Mortality Epilepsy Network and control patients from the French national research database of epilepsy monitoring units [40]. Patients with drug-resistant focal epilepsy were eligible. Sixty-two SUDEP cases and 620 controls were included. Of 21 potential risk factors, 7 were selected, and a score ranging from –1 to 8 was developed (Table 4), with high

Table 2
SUDEP-7 inventory [38].

SUDEP risk factors	Weighting	Remarks
1 >3 tonic-clonic seizures in last year	0 or 2	
2 >0 tonic-clonic seizure(s) in last year	0 or 1	If factor 1 present score as 0
3 >0 seizure(s) over the last year	0 or 1	If factor 4 present score as 0
4 >50 seizures of any type over the last year	0 or 2	
5 Duration of epilepsy >30 years	0 or 3	
6 Use of >2 antiseizure medications	0 or 1	
7 Developmental disability, I.Q. <70 or too impaired to test	0 or 2	
Total score	0 to 10	

Table 3
SUDEP-3 inventory [39].

SUDEP risk factors		Weighting
1	Generalized tonic-clonic seizure frequency >3in last year	0 or 1
2	Seizure of any type >0 in last year	0 or 2
3	Intellectual disability	0 or 1
Total score		0 to 4

Table 4
SUDEP-ClinicAI Risk score (SUDEP-CARE) [40].

SUDEP risk factors		Weighting	Remarks
1	Generalized tonic-clonic seizures in last year: between 1 per month and 1 per year	0 or 1	If factor 2 present score as 0
2	Generalized tonic-clonic seizures in last year: >1/month	0 or 1	
3	Presence of intellectual disability	0 or 1	
4	Current or past depressive disorder	0 or 1	
5	Respiratory symptoms during or after seizure	0 or 1	
6	Sleep-related or nocturnal seizures	0 or 1	
7	Able to alert someone of an oncoming seizure	0 or -1	
8	Seizure-related falls	0 or 1	
Total score		-1 to 8	

discrimination capabilities (area under the receiver operating curve 0.85, 95 % CI: 0.80–0.90). The threshold of 3 combined a good sensitivity (82.3 %, 95 % CI: 72.7 %–91.8 %), with a good specificity (82.7 %, 95 % CI: 79.8 %–85.7 %).

3.2.4. *Kempenhaeghe SUDEP risk score*

In the Academic Center for Epileptology Kempenhaeghe (Heeze, The Netherlands), one co-author (JA) developed an evidence-based SUDEP risk score in 2011 [6,41]. This SUDEP risk score ranges from 0 to 8 and is calculated based on the presence of 8 items where each item resulted in 1 point (Table 5) [42]. This SUDEP risk score was used both to assess the need of nocturnal surveillance with a seizure detection device in residential patients and in the ECOMRAID trial (Epileptic COMplication RATE in patients with Intellectual Disability) [42]. However this score was not validated.

3.3. *Design and validation of SUDEP risk scores*

3.3.1. *Design*

The design of the different SUDEP risk scores is compared in Graph 2. Each SUDEP risk score includes the frequency of GTC. “SUDEP-3” and “SUDEP-7” scores may also increase according to the frequency of other seizure types [38]. Overall, the maximum proportional contribution of the frequency of GTC in maximal SUDEP risk scores, varies between 25 and 75 %. Since “the frequency of GTC” and “the lack of nocturnal

Table 5
Kempenhaeghe SUDEP risk score [42].

SUDEP risk factors		Weighting	Remarks
1	Preceding year: >0 generalized tonic-clonic seizure a year	0 or 1	Score 1 if score item 2 is 1
2	Preceding year: >0 generalized tonic-clonic seizure a month	0 or 1	Score 1 if score item 3 is 1
3	Preceding year: >0 generalized tonic-clonic seizure a week	0 or 1	
4	Age <45 years	0 or 1	
5	Duration of epilepsy >10 years or onset age of <10 years	0 or 1	
6	Male sex	0 or 1	
7	Sleeping alone	0 or 1	
8	Often sleeping in prone position	0 or 1	
Total score		0 to 8	

surveillance/sleeping alone” are major risk factors for SUDEP [7,8], and since their combined presence even increases the risk for SUDEP [31], obviously they should have enough weight in a SUDEP risk score.

The SUDEP-CARE score and the Kempenhaeghe SUDEP risk score are also influenced by seizure characteristics and conditions that can increase (“respiratory symptoms during or after seizure”, “sleep-related or nocturnal seizures”, “sleeping alone”, “often sleeping in prone position”, “seizure-related falls”) or reduce the risk (“able to alert someone of an oncoming seizure”) [40,43].

The factors of “intellectual disability” and (longer) “duration of epilepsy” may be associated with refractory epilepsy.

“Intellectual disability” was associated with increased risk for SUDEP in the AAN guideline on SUDEP (low evidence) [7,8], but this was not confirmed in the Swedish population-based study [31]. We were therefore surprised about its proportional weight in these scores (2 out of maximal 10 points in the “SUDEP-7” risk score [39,44–47] and 1 out maximal 4 points in the “SUDEP-3” risk score [39]).

Finally, the different scores also contain some other factors, which are probably associated with the specific population (“current or past depressive disorder”) [40] or literature source (“age <45 years”, “male sex”, “onset age of <10 years”) [6,41].

The proportions mentioned (e.g., of GTC frequencies) apply when the SUDEP risk score is maximal. In “real life” they can be higher or lower depending on the presence or absence of other risk factors in the score. In the ECOMRAID trial, the correlation between the GTC frequency and the Kempenhaeghe SUDEP risk score was high (rho 0.83) although the proportion of the GTC frequency with a maximum Kempenhaeghe SUDEP risk score is only 37.5 % [42].

The use of more than 2 antiseizure medications increases SUDEP-7 risk score. Although previously considered as a risk factor for SUDEP, polytherapy was associated with a reduction of the risk for SUDEP in the Swedish study (OR 0.31, 95 % CI: 0.15–0.66) [31].

The variables of a SUDEP risk score must be clearly definable without interobserver variability. Observations by witnesses are probably not always reliable (e.g., respiratory symptoms during or after a seizure in the “SUDEP-CARE”-score) [40].

SUDEP risk scores need updates based on the results of systematic reviews and relevant studies.

3.3.2. *Validation*

SUDEP risk scores need also external validation. The Kempenhaeghe SUDEP risk factor score [42] has not yet been validated in contrast to the SUDEP-7 inventory [38], the SUDEP-3 inventory [39], or the “SUDEP-CARE”-score [40]. Because the incidence of SUDEP is low, the number of included patients has to be high enough to find statistical relevant results for the validation of the score.

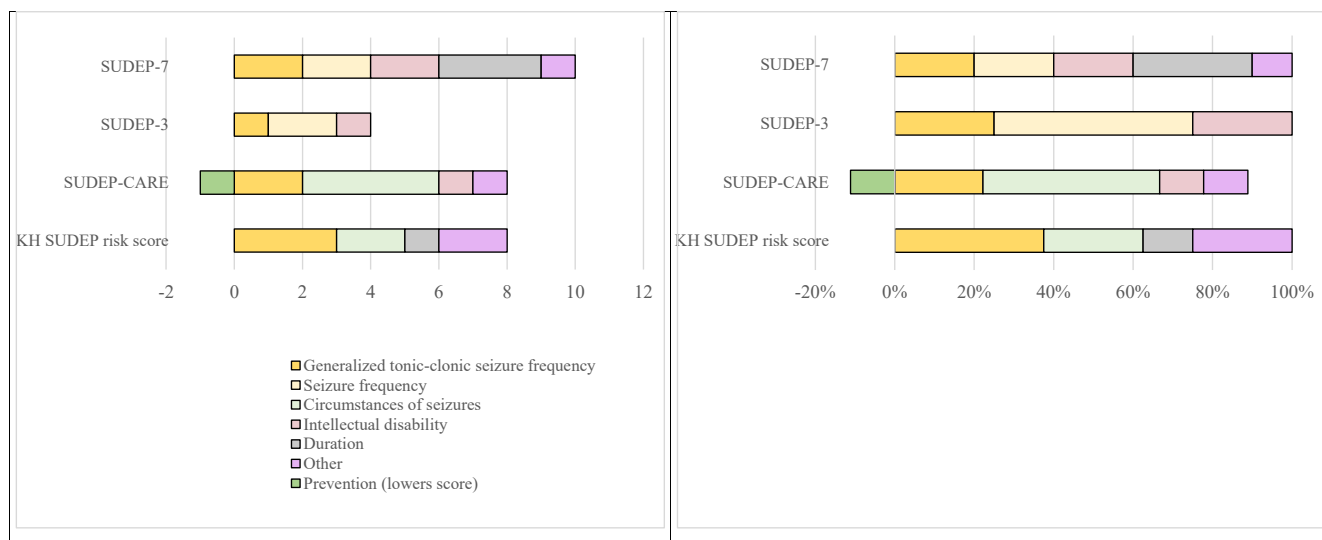
We recommend the validation of a SUDEP risk score in a population representative for all epilepsy patients, or at least for a defined subpopulation. Some current SUDEP risk scores were validated in a population from an epilepsy monitoring unit [39,40]. Such patients are a highly selected group, often candidates for epilepsy surgery. This might result in a higher number of cases at young age, suffering from focal seizures predominantly from the temporal lobe. However, people with intellectual disability and drug-resistant epilepsy are often poor candidates for epilepsy surgery because of difficulties to localize the seizure onset zone or because of the presence of multifocal seizure-onset zones. As a result, they might be underrepresented in such population.

4. *Hypothetic mechanisms of SUDEP*

In this part of the manuscript, we aim to review different hypotheses on the mechanism of SUDEP, based on animal or human studies.

4.1. *Ictal asystole*

Initially it was assumed that ictal asystole would be the mechanism



Graph 2. Design of SUDEP risk score: points (left) and proportions (right): [38–40,42]. SUDEP-CARE: SUDEP-Clinical Risk score, KH: Academic Center for Epileptology Kempenhaeghe, Heeze, The Netherlands, SUDEP: sudden unexpected death in epilepsy patients.

of SUDEP. Ictal bradycardia is considered a rare event, which affects <5 % of epilepsy patients [48,49]. Asystolic episodes following ictal bradycardia seem to occur even less frequent (0,3–0,4 %) [50,51].

However, in the MORTEMUS-study (Mortality in Epilepsy Monitoring Unit Study), video EEG with cardiorespiratory data, available for 9 cases of SUDEP, showed a consistent and previously unrecognized pattern after a GTC. Tachypnea (18–50 breaths per min) was followed within 3 min by transient or terminal cardiorespiratory dysfunction. Where transient, the dysfunction later recurred with terminal apnea occurring within 11 min of the end of the seizure, followed by cardiac arrest [52]. In all 9 cases, terminal apnea was observed before terminal asystole. Eight of the 9 patients were found in prone position (head in the pillow). Simultaneously, flattening of electrical brain activity was observed.

There are some major differences between ictal asystole and SUDEP. In SUDEP asystole was a postictal phenomenon after a GTC, always preceded by apnea and by a diffuse EEG suppression (“electrical shutdown”) [52,53]. In ictal asystole, a focal (mostly temporal) ictal discharge without obligatory generalization leads to bradycardia and asystole of duration range 3–20 s (although longer pauses have been observed) with diffuse EEG suppression (as a consequence of a reversible cerebral hypoperfusion) [54] but usually without the need of resuscitation, even with prolonged ictal asystole [50,51,55–58].

4.2. Serotonin and adenosine hypotheses of SUDEP

The parallel respiratory and cardiac dysfunction suggests a central dysfunction of brainstem centers that are involved in the control of respiration and heart rhythm [52]. Research in SUDEP animal models has led to the serotonin (5-HT) and adenosine hypotheses of SUDEP [59].

A GTC causes release of neurotransmitters including adenosine. Activation of presynaptic A1-receptors by adenosine leads to seizure termination [59,60]. However excessive adenosine signaling can also cause adverse consequences (e.g., respiratory depression, postictal generalized EEG suppression (PGES), neuronal dysfunction following spreading depolarization) and activation of A2a-receptors exerts mostly proconvulsive effects [59,61,62]. Adenosine induces respiratory depression by inhibitory actions on the neurons in the rostral ventral lateral medulla (RVLM) which controls the rhythm and rate of respiration. This diminishes the central drive to the lungs and causes apnea [59].

The apnea causes elevated pCO₂. 5-HT neurons (located around the basilar and other large arteries and other large arteries in the medulla) are strongly stimulated by a small drop in pH, indirectly reflecting CO₂ concentration, which activates brainstem raphe neurons to increase the release of 5-HT [59]. Periaqueductal gray (PAG) neuron express several 5-HT receptor subtypes and their activation by 5-HT triggers autorresuscitation mechanisms [59] (triggering of gasping behavior in response to apnea) and other restorative respiratory response mechanisms (hypercapnic and hypoxic ventilatory response) [63]. This effect of serotonin can potentially compensate, in part, for the adenosine-mediated respiratory depression [63]. If these mechanisms are not sufficient to overcome the apnea, it may result in SUDEP. Seizures may reach central structures controlling breathing either by direct synaptic connections from cortex to forebrain structures (such as amygdala) and brainstem, or by a travelling wave of depolarization inhibiting everything on his path [63].

4.3. Postictal generalized EEG suppression

Prolonged postictal generalized EEG suppression (PGES) consists of the immediate postictal (within 30 s of seizure cessation) generalized reduction of the cerebral EEG activity below 10 μ V in amplitude and minimum duration of 1 s [26].

PGES was observed in several cases of SUDEP and near SUDEP [26,30,64] and has been proposed as an electrophysiological marker for SUDEP, but its significance is yet to be defined [34,65]. Some SUDEP risk factors, such as GTC frequency [66], are strongly associated with PGES.

Some hypotheses have been postulated about the triggers of PGES. First, PGES may reflect brainstem spreading depolarization, since PGES is associated with GTC and particularly with the occurrence of symmetric tonic arm extension and oral tonicity [26,67]. Second, it may passively arise from cerebral cortical neuronal exhaustion resulting from the depletion of energy substrates after prolonged and hyper-synchronous neuronal excitation or the decreased cerebral blood flow from postictal cardiorespiratory dysfunction [65,67]. Finally, PGES may represent the activation of a reticular-thalamo-cortical inhibitory network responsible for switching off cortical activity and seizure termination [67].

4.4. Hypotheses for other concomitant mechanisms

Bradycardia and asystole may be secondary to hypoxemia and hypoventilation. Prolonged apnea activates carotid chemoreceptors causing arousal and eventually vagally mediated bradycardia or even cardiac arrest.

In SUDEP, the timing of the terminal apnea precedes asystole (despite earlier nonterminal cardiac rate and rhythm changes). The reason could be that in the heart, unlike the lungs, intrinsic cardiac pacemaker activity, which allows resilience of cardiac function despite disordered central influence, may initially compensate by creating a margin of cardiac function in seizures with cardiac and respiratory dysfunction [59].

Also, concomitant obstructive mechanisms such as tonic activation of diaphragm or laryngospasm may contribute to the seizure-related apnea.

4.5. The epileptic heart

Chronic cardiac dysfunction in people with epilepsy might also contribute to the mechanisms of SUDEP [34,68]. The term “epileptic heart” refers to the damage of heart and coronary vasculature resulting from chronic epilepsy and leading to mechanical dysfunction and electrical instability [69]. Pathologic studies in chronic epilepsy patients, showed myofibril damage, extracellular matrix deposition, and myocardial fibrosis probably due to recurrence of catecholamine stimulation [67]. Interstitial myocardial fibrosis and myocyte hypertrophy are the most frequent pathological abnormalities among SUDEP cases [67]. Myocardial fibrosis predisposes to arrhythmias such as atrial fibrillation, supraventricular tachycardia, bundle branch block, ventricular, or premature atrial depolarizations [70]. The definition of SUDEP excludes known causes of mortality as cardiac comorbidities and sudden cardiac death, although chronic epilepsy itself predisposes to heart disease and increases the risk of sudden cardiac death [69]. This mechanism of death should be differentiated from SUDEP, but it may make the patient more vulnerable to SUDEP in the postictal phase of a GTC.

4.6. Genetic variants

Some genetic variants might be responsible for SUDEP, in particular those responsible for epileptic encephalopathies, e.g. mutations in the voltage-gated sodium channel gene SCN1A in Dravet syndrome. However, disentangling the direct role of the mutation from that of refractory seizures in promoting SUDEP remains difficult [34].

Two types of cardio-cerebral channelopathies exist. In inherited cardio-cerebral channelopathies inherited mutations in ion channels expressed in brain and heart generate a dual seizure and arrhythmia phenotype with both aberrant neuronal and cardiac excitability. As an example, long QT syndrome (LQTS) is characterized by ventricular arrhythmias that lead to syncope or sudden cardiac death but some genetic variants causing LQTS might also cause epilepsy [34,71]. In an acquired cardiac channelopathy, (chronic) epilepsy overstimulates sympathetic signaling pathways in cardiomyocytes and secondarily alters the gene and protein expression of cardiac ion channels in association with proarrhythmic abnormalities in cardiac conduction [71]. The contribution of an acquired cardiac channelopathy to the elevated risk of sudden death in epilepsy is however still unclear [71].

5. Seizure detection devices

5.1. Aims

Seizure detections devices (SDDs) in general target continuous, unobtrusive and relatively inexpensive monitoring of predefined seizure types. SDD should also respect the independence and privacy of the

patient.

A joint working group of the ILAE and the International Federation of Clinical Neurophysiology (IFCN) developed a clinical practice guideline on automated seizure detection using wearable devices, recommending “using clinically validated wearable devices for automated detection of tonic-clonic seizures (TC) when significant safety concerns exist, especially in unsupervised patients who do not share a bedroom but where alarms can result in rapid intervention within 5 min” [72,73]. However, although SUDEP poses the most impressive risk, other (and more frequent) seizure emergencies (e.g., resulting in serious seizure-related injuries and seizure-related respiratory complications) [42] should also be detectable.

5.2. Individual sensors, multimodal seizure detection and video detection systems

Commonly employed sensors for physiological signals for seizure detection include: electroencephalography (EEG), electrocardiography (EKG), plethysmography (PPG), blood oxygen saturation, accelerometry (ACM), electrodermal activity (EDA), electromyography (EMG), and an indirect body movement mattress sensor [74,75]. EEG measures the change of electrical impulse in the brain [74,75]. EKG measures the electrical activity of the heartbeat [74,75]. ACM quantifies movement changes in velocity and direction [74,75]. EDA measures electrical conductance of the skin in response to sweating [74,75]. EMG implies the measurement of muscle activity (tonic phase of seizures) [74–77]. A mattress sensor detects clonic movements. An under-mattress Electro-Mechanical Film (Emfit®) is triggered by rhythmic motor activity of a specifiable duration, frequency, and intensity using a quasi-piezoelectric material sensitive to changes in mattress pressure [78]. Respiratory function (depressed postictal blood oxygen saturation) is also used for seizure detection with fairly good results [79]. However the obtrusiveness and false-positive responses make it less useful for long-term detection at home. PPG utilizes a light-approach to sense the rate of blood flow [74,75] and also has the potential to detect breathing rate [80]. Automated video detection systems use different parameters (e.g., velocity, area, duration, rotation, oscillation, angular speed, displacement) for seizure detection [81–85].

The use of multiple sensors may allow to increase the sensitivity and decrease the false detections [86].

5.3. Standards for testing and clinical validation

Currently various SDDs exist [87,88]. Beniczky and Ryvlin have proposed standards for testing and clinical validation of SDDs. Five phases of seizure detection studies are based on key features (subjects, recordings, data analysis, alarms and reference standards) and outcome measures [89]. Depending on how the studies address these features, they are classified into phases 0–4 [89]. The final confirmation of the safety and accuracy of a SDD is provided in a phase-3 study [89]. Such study must be prospective, multicenter, enrolling at least 20 patients with at least 30 seizures (for a sensitivity over 95 %), with real-time detection of seizures (signal analysis continuously running during the recording) using a pre-defined algorithm and detection-threshold with the reference standard from video or video-EEG recordings interpreted by experts [89]. Phase-4 studies are open-label, follow-up studies addressing aspects related to the feasibility and the utility of the devices in the home environment of the patients (in-field studies) [89] enrolling at least 50 patients with at least 75 seizures (for a sensitivity over 95 %). Analysis and seizure detection need to be prospective and real time [89].

5.4. Phase-3 and phase-4 studies

The ILAE-IFCN working group clinical practice guideline was based on a systematic literature review available on October 16, 2019 using the search string: ((automated detection) OR (algorithm AND detection)

OR (wearable AND detection)) AND (epilepsy OR seizure) [72,73]. Meritam Larsen et al. updated the search in PubMed up to August 2023 [90] using the same search string as used by the ILAE-IFCN working group. They found six phase-3 studies [77,91–95] and four phase-4 studies [92,95–97]. One study reported both phase 3 and 4 [92].

Five different SDDs are analyzed in phase-3 studies, of whom 3 also in phase-4 studies. All but 1 [95] were “wearables”. **Epicare®** is a wrist-worn, wireless accelerometer sensor for detecting GTCs [91]. **Seizure link®** is a wearable surface EMG device placed on the brachial biceps muscles [77]. **NightWatch** is a bracelet worn at night at the upper arm and contains a photoplethysmographic heart rate module and a 3-dimensional accelerometer [92,93]. **Empatica** is a wrist-worn device combining accelerometers (ACM) and electrodermal activity (EDA) sensors [94]. **Nelli®** is an automated detection system of nocturnal motor seizures using audio–video streaming, computer vision and an artificial intelligence-based algorithm [95].

The characteristics of these 5 SDDs are summarized in Table 6. The following data were extracted from the studies: 1) (commercial) name of SDD; 2) signal used for seizure detection; 3) number and age of included patients; 4) seizure-types that were analyzed; 5) sensitivity; 6) false alarm rate (FAR); 7) latency of seizure detection. A quantitative synthesis (meta-analysis) was not possible, due to the large heterogeneity in study design and the use of different devices and algorithms. Hence, we conducted a qualitative synthesis of the included studies.

All devices detected GTC. Van Andel et al. introduced the concept of major seizures as a representation of clinical urgency of seizures [76]. Major seizures are specified by medical professionals according to seizure semiology, which includes loss of consciousness, falls, long duration, intense movements, respiratory problems, cluster or status, salivation/vomiting, oxygen drop, migraine, or visual problems [98]. Four types were denoted such: TC because of the increased risk for SUDEP [7,8], prolonged generalized tonic (GT, >30 s) because they may cause respiratory distress, focal hyperkinetic (HK) given the high risk of injury and other major (OM) consisting of TC-like seizures with atypical semiology and clusters of minor seizures lasting >30 min. All 3 seizure types (TC, GT, HK) and the less well-defined class OM are characterized by at least 1 of these elements [92].

All 5 SDDs have high sensitivity for TC combined with an acceptable low positive false alarm rate. As stated earlier, sensitivity must always be seen in relation to the false positive alarm rate. The sensitivity ranged from 90 % to 100 %. Some studies reported the sensitivity for the total of the relevant seizures, while others reported the median SDD sensitivity per patient. We propose to report the latter because this allows to avoid the bias resulting from dysfunction or inappropriate use of SDD by particular patients and an overrepresentation of persons with many seizures.

The false alarm rate (FAR) was reported either by hour or by night. Although the figures do not vary widely, they should still be interpreted with some caution. Most authors indicate the false positive alarm rate (FPAR), referring to the frequency of alarm in the absence of TC or other major seizure. In phase-3 studies of NightWatch [92] also the false negative alarm rate (FNAR) was indicated, referring to the failure of detecting a seizure. The FPAR may be lower in a study design with detection of major [92,93] or hypermotor seizures [95] besides only TC [77,91], since some alarms may be classified as a correct seizure detection instead of a type I error (false positive). On the other hand, in the phase-3 studies of NightWatch [92] minor seizures were considered false-positive results, as opposed to the phase-3 Nelli® study [95]. In a phase-3 study in children the high FAR was largely attributed to the heart rate part of the algorithm and reduced by 60 % when the alarm was triggered only when the patient was in a horizontal position (from 0.2/h to 0.08/h), but not at the expense of sensitivity (roughly the same and comparable to that in the adult population) [93].

In all phase-3 studies conducted in epilepsy monitoring units, the gold standard was based on expert evaluation of long-term video electroencephalography. In the NightWatch phase-3 studies, conducted in

residential care institutes, the gold standard was the expert evaluation of long-term video but without simultaneous electroencephalography. This was inevitable because of the long duration of monitoring (3 months) but seemed justified based on the overall interobserver agreement of 0.77 (95 % CI: 0.65–0.89).

Four phase-4 studies demonstrated the feasibility of wearable SDDs and their usability for detecting GTCs in the home environment of the patients. However, most patients included into the phase-4 studies had severe epilepsy and intellectual disability, living in a residential care setting.

6. Discussion

The main risk factors for SUDEP are the presence and frequency of GTC and the lack of nocturnal supervision. Other but less important risk factors refer to insufficient seizure control (“not adding antiseizure medication while still seizures”, “any seizure type present”) or to the circumstances in which the patient is at risk for SUDEP (positive risk factors such as “prone position”, “being in bed”, “being in the bedroom” and the negative risk factor “use of listening device”).

In the attempt to prevent SUDEP, it is important to recognize patients at increased risk for SUDEP. Despite the progress in knowledge about the individual risk factors for SUDEP, the impact of the combined presence of risk factors is largely unknown. Four SUDEP risk scores are available: SUDEP-7 inventory, SUDEP-3 inventory, SUDEP-CARE score and Kempenhaeghe SUDEP risk score. The design of the scores is different, resulting in a different weight and proportion of (strong) risk factors. In a given SUDEP risk score, the proportional impact of the different risk factors may change in the case not all risk factors that could determine the score are present. Depending on the population, the weight of GTC in the final score may differ. All but the Kempenhaeghe SUDEP risk score were validated but all validations were subpopulation linked. Although the previously mentioned studies attempted to recognize independent risk factors, the risk may additionally increase or decrease when 2 or more risk factors are simultaneously present (E.g., a 67-fold increased risk of SUDEP for the combination of having at least one GTC and not sharing a bedroom with someone).

In the MORTEMUS study, a long-lasting period of central apnea after a GTC led to a terminal asystole in the fatal cases. Nocturnal supervision reduces the risk of SUDEP with an OR up to 0.4. Five clinically validated SDDs showed high sensitivity for TC combined with low FPAR (EpiCare®, Seizure Link®, NightWatch, Empatica and Nelli®). The ILAE-IFCN guideline recommends their use in unsupervised patients with TC who do not share a bedroom but where alarms can result in rapid intervention within 5 min”.

7. Perspectives

SUDEP remains a rare occurrence and SUDEP incidence still appears to be much lower than the incidence of the combination of still having GTC and sleeping alone [31]. Many questions on SUDEP risk factors remain unanswered. The AAN estimates (a 6 times lower incidence of SUDEP in children) were initially contradicted by the results of a Swedish [31] and a Canadian study [33] but confirmed in a later Danish study [34]. The reason for this discrepancy is still unclear. Both Scandinavian studies [31,34] were nation-wide population-based studies, and they have the same figures for adults. In our opinion, there is a need for a new systematic review with updates on a regular basis. Ideally, the initiative and responsibility for updating is taken by the ILAE. Also, a validated SUDEP risk score system that clearly shows which population it applies to, would be of great added value both to inform patients and for risk stratification in healthcare. SUDEP risk scores need updates based on the results of additional systematic reviews and relevant studies.

Also, in terms of the mechanism of SUDEP, some questions remained unsolved so far. For example, postictal generalized EEG suppression

Table 6

Studies analyzing the technical performance and diagnostic accuracy of real-time tonic-clonic seizure detection wearables: h: hour, y: year, EMU: epilepsy monitoring unit (long-term video-EEG monitoring), sEMG: surface electromyography, GTC: generalized tonic-clonic seizure, 3D-A: 3D- accelerometer, 95% CI 95% confidence interval, RCI: residential care institute, GTC: generalized tonic-clonic seizure, * major motor seizures = (1) TC: tonic clonic, GT: long generalized tonic (>30s), HK: focal hyperkinetic, OM: other major consisting of TC-like seizures with atypical semeiology and clusters of minor seizures lasting >30 minutes, HRP: heart rate photoplethysmography, FPAR: false positive alarm rate, FNAR: false negative alarm rate, ACM: accelerometer, EDA: electrodermal activity, IQR: interquartile range. [77,91–96,99].

Device and phase	Sensor	Population	Seizure type detected	Device sensitivity	False alarms	Alarm time from seizure onset	
WEARABLE SEIZURE DETECTION DEVICES							
Epi-Care® (Phase 3)	Wrist 3D-A	EMU 73 patients (of whom 20 with GTC) Age: Range: 13–63y Mean: 37y 4878h	GTC	Seizures: 35/39 = 89.7% Per patient: Mean 91% 95% CI: 80%–100% Range: 0%–100%	0.2/24h 0/24h (night)	GTC: Mean 33s Focal seizure: Mean: 55s Range: 38s–73s	Beniczky et al. [91]
(Phase 4)		71 patients Age: Range: 7–72y Median: 27y	GTC	Median: 90% Mean: 85% Range 0–100%	Median 0.1/24h Mean 1.4/24h Range 0–12/24h	In field study: not applicable	Meritam et al. [96]
Seizure Link® (Phase 3)	Above brachial biceps muscle sEMG	EMU 71 patients (of whom 20 with GTC) Age: Range: 10–62y Mean 34y Median: 34y 3735.5h	GTC	Seizures: 30/32 = 93.8% 95% CI: 86%–100%) Per patient: Mean 95% Median 100% (range 50–100%)	0.67/24h	9s Range: –4s–48s (from start of tonic phase)	Beniczky et al. [77]
NightWatch (Phase 3+4)	Upper arm HRP+3D-A	RCI (video monitoring) 28 patients Age: Range: 15–67y Mean: 29y 1826 nights	Major → TC → GT → HK → MO →	Up to 86% (per patient) 95% CI: 77%–93% 77% (seizures) Median: 96% (per patient) 95% CI: 80%–100% 81% (seizures) Median: 89% 95% CI: 33%–100% Median: 73% 95% CI: 50%–100% Median: 84% 95% CI: 50%–100%	FPAR: 0.25/night 95% CI: 0.04–0.35 FNAR: 0.03/night 95% CI: 0.01–0.05	Not specified. Seizure considered detected if within 3min before and 5min after onset.	Arends et al. [92]
(Phase 3)		RCI (video monitoring) 23 patients Age: Range: 3–17y Mean: 9.7y 497 nights	Major → TC →	Seizures: 305/384 = 79.4% Per patient: Median: 93.2% Mean: 85.9% 95% CI: 59.9%–100% Range: 47.4%–100% Seizures: 92/103 = 89.32%	FPAR: 0.08/h FNAR: Median: 0.02/h (night) Range: 0.003–0.12	7s	Lazeron et al. [93]
(Phase 4)		In-home 532 nights	Major motor*	Up to 100%	0.04/h (night)		van Westrhenen et al. [97]
Empatica (Phase 3)	Wrist ACM+EDA	Multicenter EMU Pediatric patients: Number: 85 Median age: 12 Range age: 6–20 Monitor: 162d Adult patients: Number: 67 Median age: 38 Range age: 21–63 Monitor: 265d	TC	Pediatric patients: 92% (95% CI: 85%–100%) Adult patients: 94% (89% CI: 89%–100%)	Pediatric patients: 1.26 (95% CI: 0.87–17.3) Adult patients: 0.57 (95% CI: 0.36–0.81)	37s	Onorati et al. [94]
	See above	In-home Age: Range: 1–82y 242 patients	GTC	Median 100% Perceived sensitivity: Median: 100% IQR: 95%–100% Perceived sensitivity: Median: 100% IQR: 95%–100% Perceived sensitivity: Median: 100% IQR: 100%–100%	Night vs day Median: 0 vs 0.29 IQR: 0–0 vs 0–0.43 Night vs day Median: 0 vs 0 IQR: 0–0 vs 0–0.29 Night Median: 0.10 IQR: 0.03–0.43		Hadady et al. [99]
NON-WEARABLE SEIZURE DETECTION DEVICES							
Nelli® (Phase 3)	Contactless audio-video signal based	EMU 51 patients Age: Range: 2–72 Median: 16 4183h (night)	Major motor	Major convulsive: 15/16 = 93.7% 95% CI: 69.8%–99.8% TC: 11/11 = 100% 95% CI: 71.5%–100% Hypermotor: 4/5 = 80% 95% CI: 28.4%–99.5%	0.16/h Median per patient 0.07/h IQR: 0–0.19	Not reported	Armand Larsen et al. [95]

(PGES) was observed in several cases of SUDEP and near SUDEP [26,30,64] and has been proposed as an electrophysiological marker for SUDEP, but its significance is yet to be defined [34,65]. Although certain genetic abnormalities are associated with a higher risk of SUDEP (e.g., gene mutations in Dravet syndrome), no gene mutation with an unambiguous link to the actual occurrence of SUDEP has yet been identified. It also remains intriguing why, both in patients in whom an ictal asystole was diagnosed and in patients who died of SUDEP, also seizures with a milder form of heart rate decrease (ictal bradycardia versus ictal systole) were reported. The SUDEP-pathway that emerges from the MORTEMUS study calls for other studies to support these observations and for further analysis of the underpinning pathophysiological pathways.

Studies should be undertaken to demonstrate the clinical value of SDDs. Because the low incidence makes it hard to use SUDEP as an endpoint, research groups might consider using surrogate endpoints as seizure-related complications.

A study is warranted to investigate to what extent SDDs save lives, increase the quality of life of patients, relieve the workload of nurses and reduce the financial costs of healthcare. SDDs may also be useful for objective quantification of seizure frequency and severity [100,101]. Both under- and over-reporting may occur and consequently affect the accuracy of therapeutic decisions in daily clinical practice. They also question the reliability of large drug studies (which are typically based on patient-reported seizure frequency) [100]. Quantification is challenging because it requires a high accuracy of seizure identification.

Theoretically, devices may allow seizure prediction prior to its clinical manifestation. A meta-analysis of pre-ictal changes in heart rate suggests that such changes are observable in some but not all individuals, but the interval between the pre-ictal changes and the start of the seizures is mostly short [102]. Further work is needed to determine in which patients heart rate monitoring would be useful for seizure prediction [102,103]. Instead of attempting to predict the exact time of an upcoming seizure, the estimation of the probability of the occurrence of a seizure within a certain time interval would be more feasible [104,105]. It is recommended that future studies with SDDs take the aspect of seizure prediction into account.

CRedit authorship contribution statement

C.P.J.A. Monté: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **J.B.A.M. Arends:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **R.H.C. Lazeron:** Writing – review & editing, Supervision. **I.Y. Tan:** Writing – review & editing, Supervision. **P.A.J.M. Boon:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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